

SEMINAR_progress_report

Rajat Ranjan Meher

Instructor:- Bhaskaran Raman

February 2026

1 February 7th - February 12th

1.1 Objective:- Read research paper

Name of paper:-Performance of Cellular Networks on the Wheels

Topics covered:-Introduction,Related work, Methodology,Network coverage, Performance, Handover , 5G applications , Conclusions

Concepts learned:-Importance of deployment policy by different operators, Factors affecting 5G performance

Notes:-Made handwritten notes of all topics

2 February 13th - February 20th

2.1 Objective:- Read research paper

Name of paper:-Analysis of 4G LTE Network Performance in Urban Areas

Findings:-

- Signal strength (dBm) directly affects call quality and throughput
- Network congestion in dense urban areas reduces performance
- Indoor environments experience weaker signals compared to outdoor

3 February 21st - February 28th

3.1 Objective:- Read research paper

Name of paper:-Analysis of 4G LTE Network Performance in Urban Areas

Findings:-

- Interference between nearby cells impacts network stability
- Performance varies across different operators and deployment strategies
- User density plays a major role in network degradation

4 March 1st - March 7th

4.1 Objective:- Read research paper

Name of paper:-Measurement Study of Mobile Network Performance in Real-world Scenarios

Findings:-

- Network performance varies significantly with location
- User experience (QoE) differs from network metrics (QoS)
- Crowdsourced data collection improves measurement accuracy

5 March 8th - March 14th

5.1 Objective:- Read research paper

Name of paper:-Measurement Study of Mobile Network Performance in Real-world Scenarios

Findings:-

- Network conditions fluctuate over time
- Large-scale data collection is required for reliable analysis
- Real-world measurements are essential for accurate performance evaluation